

Dr. Rohitkumar R. Upadhyay

Applied Cryptographer, QNu Labs



About Me

I am an applied cryptographer with a Ph.D. in Mathematics, specializing in privacy-preserving cryptography (FHE, MPC, ZKPs), post-quantum cryptography (PQC), and secure implementations. At [QNu Labs](#), I develop and analyze high-assurance post-quantum cryptographic systems, focusing on protocol efficiency and side-channel resilience. I earned my Ph.D. from [MNNIT Allahabad](#) (2022–2025) and was selected as one of three global laureates for the Post-Quantum Algebraic Cryptography trimester at the [Institut Henri Poincaré](#), Paris. I previously served as Assistant Professor of Mathematics at [Vaze College](#), Mumbai (2019–2022), combining research, industry, and teaching to advance practical post-quantum cryptography.

Education

- 2022 – 2025 **Ph.D. in Mathematics**, [MNNIT Allahabad](#), India
- Thesis: *A Study of Secure Computation using Post-Quantum Cryptography*
 - Advisor: [Dr. Sahadeo Padhye](#)
 - Proposed efficient dynamic multi-party FHE frameworks based on **GSW** and **FHEW** for scalable secure computation.
 - Designed **FHE-Pro**, integrating proxy re-encryption with FHE to reduce communication overhead and enable non-interactive key generation.
 - Developed verifiable FHE protocols using **SNARKs** and optimized homomorphic computation for efficient privacy-preserving machine learning.
 - Explored **VOLE-in-the-Head** techniques for efficient post-quantum digital signatures.
- 2017 – 2019 **M.Sc. in Mathematics**, [University of Mumbai](#), India
- 2014 – 2017 **B.Sc. in Mathematics**, [NES Ratnam Degree College](#), Mumbai, India

Experience

- Apr 2026 – Present **Applied Cryptographer**, [QNu Labs](#), Delhi, India
- Jun 2025 – Mar 2026 **Cryptography Analyst**, [QNu Labs](#), Bangalore, India
- Developing **algorithmic optimizations** and **formal security analyses** for constant-time ML-KEM and ML-DSA to ensure resistance against physical leakages.
 - Formulating **theoretical leakage models** and designing robust masking and blinding countermeasures for post-quantum cryptographic primitives.
 - **Designing and formally modeling** efficient MPC and FHE-based protocols for secure data sharing, focusing on reducing communication and improving scalability.
 - Evaluating NIST PQC standards and global benchmarks to inform internal transition strategies.

- Sep 2024 – Dec 2024 **Trimester Laureate**, *Institut Henri Poincaré (IHP)*, Paris, France
 Selected as one of three global laureates for the IHP Trimester on Post-Quantum Algebraic Cryptography; worked with Prof. [Ludovic Perret](#) on MPC-in-the-Head-based signatures.
- 2019 – 2022 **Assistant Professor**, *KET's Vinayak Ganesh Vaze College*, Mumbai, India
 o Taught undergraduate courses including Topology, Number Theory, and Cryptography.

Technical Skills

FHE/MPC Libraries	OpenFHE , SEAL , Lattigo , libsark , Circom , MP-SPDZ
PQC & SCA	PQClean , liboqs , ChipWhisperer , constant-time coding, masking/blinding
Programming	C/C++ (Advanced), Python (Advanced — SAGE, NumPy), Go (Intermediate)

Research Publications

- 2025 **Upadhyay, R.R., Padhye, S., Sahu, R.A., Saraswat, V.** *Towards Privacy and Integrity: SNARK-Driven Verifiable FHE for Outsourced Computation*. In: **ACNS 2025 Workshops**. Springer LNCS.
- 2025 **Upadhyay, R.R., Padhye, S., Saraswat, V.** *A Non-Interactive and Dynamic Multi-Party FHE Scheme Based on FHEW*. **Peer-to-Peer Networking and Applications**. (SCIE).
- 2025 **Upadhyay, R.R., Padhye, S.** *A Non-Interactive and Secure Dynamic Multi-Party FHE Scheme*. **International Journal of Information Security**. (SCIE).
- 2025 **Kumar, R., Dabral, A., Upadhyay, R.R.** *Cryptanalysis of a Lattice-Based Blind Ring Signature Scheme*. **National Academy Science Letters**. (SCIE).
- 2025 **Upadhyay, R.R., Padhye, S.** *SoK: Digital Signatures from VOLEith*. **Human-2025 Symposium**. (Best Paper Award).
- 2025 **Upadhyay, R.R., Padhye, S.** *Privacy-Preserving Image Retrieval Based on Threshold Secret Sharing*. **Human-2025 Symposium**.
- 2024 **Upadhyay, R.R., Padhye, S.** *Efficient and Secure MPC Through Integration of FHE and Proxy Re-Encryption*. **SN Computer Science**.
- 2024 **Upadhyay, R.R., Padhye, S.** *Advancements in FHE over the Integers: A Comprehensive Survey and Analysis*. **Surveys in Mathematics and its Applications**.
- 2023 **Upadhyay, R.R., Padhye, S.** *Enhancing FHE Over the Integers: Beyond Binary Numbers and Batch Processing*. **ICCNMML 2023**. Springer LNNS.
- 2023 **Upadhyay, R.R., Padhye, S.** *Multi-Key Fully Homomorphic Encryption Scheme Over the Integers*. **REDCYSEC 2023**. Springer LNNS.

Selected Invited Talks and Panels

- Jun 2026 *Post-Quantum Cryptography Across the Data Lifecycle: In Transit, At Rest, and In Computation* — **ICMMICSD-2026**, [VIT Vellore](#).

- Apr 2026 *A Gentle Introduction to Modern Cryptography* — Alumni Lecture Series, [University of Mumbai](#).
- Sep 2025 Expert lectures ($\times 3$) on cryptography careers, statistics in PQC, and discrete math approaches to PQC — [VIT Vellore](#).
- Jul 2025 Panelist, “PQC Transition Strategy” — [SETS](#), Chennai.
- Jun 2025 *Towards Privacy and Integrity: SNARK-Driven Verifiable FHE for Outsourced Computation* — [PrivCrypt 2025](#) (ACNS), Munich.
- Mar 2025 *SoK: Digital Signatures from VOLEitH* — Human-2025, Kalyani (**Best Paper Award**).
- Sep 2024 Secure Computation — Introductory Summer School, IESC Cargèse, Corsica.
- Jul 2024 *Secure Computation* (FHE, MPC, and hybrid FHE-MPC) — [NIWC 2024](#), MNNIT Allahabad.
- May 2024 *Secure Multi-Party Computation: A Gentle Introduction* — SparQ series, [QNu Labs](#).
- Dec 2023 Secure computation with Fully Homomorphic Encryption — [CIMPA Research School](#), Pondicherry University.

Awards & Grants

- June 2025 **Financial Support for PrivCrypt 2025**, Supported by **CIMPA, CRSI, and BOSCH**, Munich, Germany
For presenting the paper *Towards Privacy and Integrity: SNARK-Driven Verifiable FHE for Outsourced Computation* at PrivCrypt 2025, a co-located workshop of ACNS 2025.
- March 2025 **Best Paper Award**, *3rd Doctoral Symposium on Human-Centered Computing (Human-2025)*
Awarded for the paper *SoK: Digital Signatures from VOLEitH*.
- March 2025 **Student Stipend (IACR)**, *Real World Crypto Symposium (RWC) 2025*, Sofia, Bulgaria
Awarded by IACR to attend the symposium and engage with the global cryptography community.
- March 2025 **Student Stipend**, *TPMPC 2025*, IISc Bangalore, India
To attend the Theory and Practice of Multi-Party Computation Workshop.
- Sep – Dec 2024 **CIMPA Fellowship / Trimester Laureate**, *Institut Henri Poincaré (IHP)*, Paris, France
Selected as one of three global **Laureates** for the *Post-Quantum Algebraic Cryptography* program.
- Dec 2023 **CIMPA Financial Support**, *CIMPA Research School*, Pondicherry University, India
For attending the Research School on Post-Quantum Cryptography.
- 2022 **CSIR-UGC JRF and NET Qualification**, [CSIR, India](#)
Qualified for Junior Research Fellowship (JRF) and Lectureship (NET).
- 2021, 2022 **GATE Qualified**, [Ministry of Education, India](#)
Graduate Aptitude Test in Engineering.